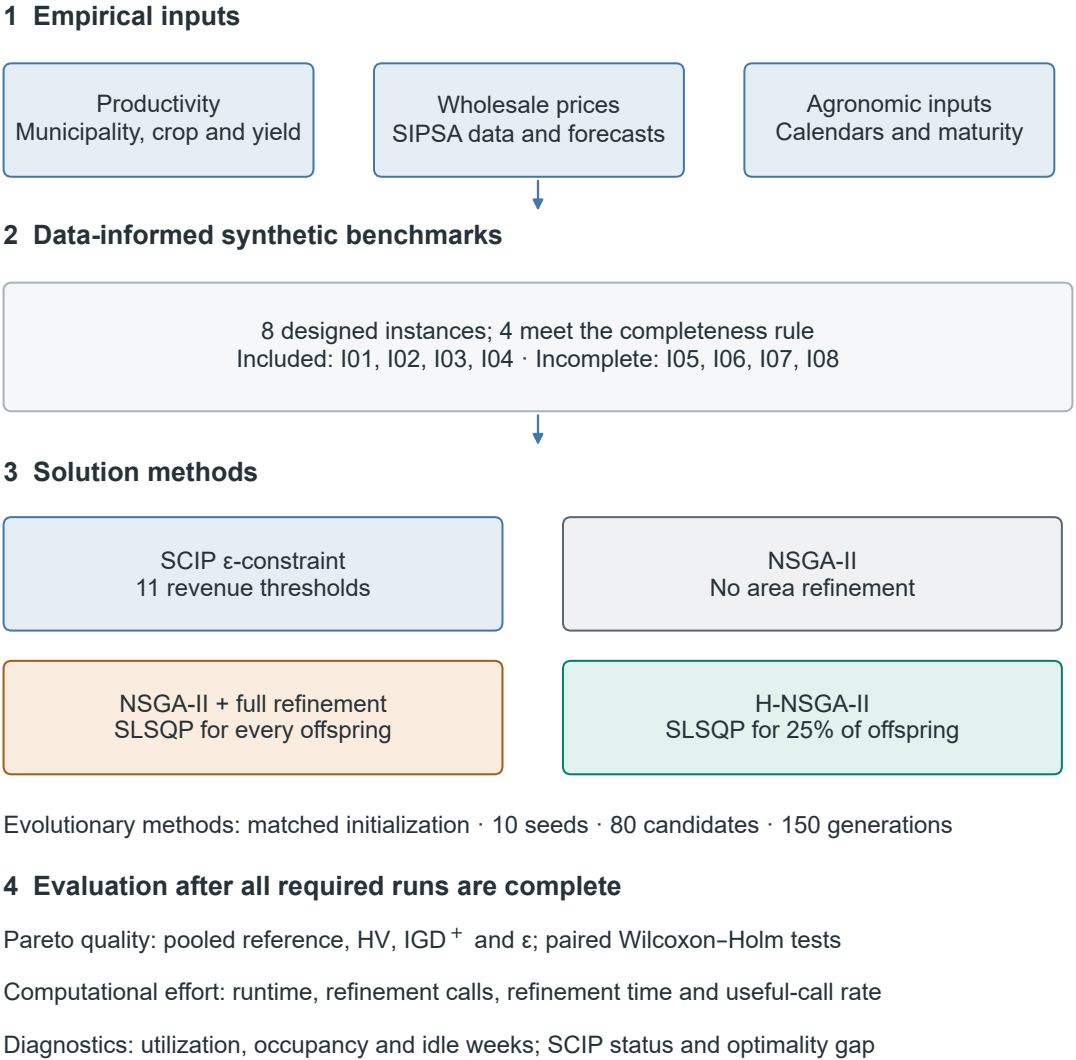

1 Empirical inputs
Productivity
Municipality, crop and yield
Wholesale prices
SIPSA data and forecasts
Agronomic inputs
Calendars and maturity
2 Data-informed synthetic benchmarks
8 designed instances; 4 meet the completeness rule
Included: I01, I02, I03, I04 · Incomplete: I05, I06, I07, I08
3 Solution methods
SCIP ε-constraint
11 revenue thresholds
NSGA-II
No area refinement
NSGA-II + full refinement
SLSQP for every offspring
H-NSGA-II
SLSQP for 25% of offspring
Evolutionary methods: matched initialization · 10 seeds · 80 candidates · 150 generations
4 Evaluation after all required runs are complete
Pareto quality: pooled reference, HV, IGD + and ε; paired Wilcoxon–Holm tests
Computational effort: runtime, refinement calls, refinement time and useful-call rate
Diagnostics: utilization, occupancy and idle weeks; SCIP status and optimality gap